

Technology Stack

Date	30 June 2026
Team ID	
Project Name	AI and Machine Learning Project
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3	Used to create a simple, responsive, and user-friendly web interface for entering applicant details and displaying the prediction result.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask handles user requests, processes input data, loads the trained machine learning model, and returns the prediction result efficiently.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset (Pandas)	The credit card approval dataset is stored in CSV format and processed using Pandas for data preprocessing and model training.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	PythonAnywhere	Used to deploy the Flask application online, making the project accessible through a live URL without requiring local installation.

5	Version Control & CI/CD	Git, GitHub	Git is used for version control, while GitHub stores the project source code, tracks changes, and enables collaboration and project submission.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Scikit-learn, Joblib, NumPy	Scikit-learn is used to train the Random Forest model, NumPy supports numerical computations, and Joblib saves and loads the trained model for prediction.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		